higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS
(Second Paper)
NQF LEVEL 3

6 NOVEMBER 2019

This marking guideline consists of 9 pages.

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✓ = 1 mark $\checkmark^{½} = \frac{1}{2}$ mark
Accept ALL mathematically valid solutions.

QUESTION 1

1.1 1.1.1

$$V = \frac{4}{3}\pi r^3$$
$$= \frac{4}{3}\pi (15)^3 \checkmark$$
$$= 14137,167 \text{ cm}^3 \checkmark$$

No marks for the formula

Do not penalise for incorrect units. (2)

1.1.2

$$V = \frac{1}{3}\pi r^2 \times h$$
$$= \frac{1}{3} \times \pi \times 15^2 \times 40 \checkmark$$
$$= 9424,778 \text{ cm}^3 \checkmark$$

(2)

1.1.3

$$l = \sqrt{h^2 + r^2} \quad \checkmark^{½}$$
$$= \sqrt{40^2 + 15^2} \quad \checkmark^{½}$$
$$= 42,72 \text{ cm} \quad \checkmark^{½}$$

1 ½ marks for the slant height.

$$\text{Area of curved surface} = \pi \times r \times l$$
$$= \pi \times 15 \times 42,72 \quad \checkmark^{½}$$
$$= 2013,133 \text{cm}^2 \quad \checkmark^{½}$$

No CA marks within this question

$$\text{Area of base} = \pi r^2$$
$$= \pi \times 15^2 \quad \checkmark^{½}$$
$$= 706,858 cm^2 \quad \checkmark^{½}$$

$$\text{Total surface area} = 2013,133 + 706,858 \checkmark^{½}$$
$$= 2719,991 \ cm^2 \checkmark$$

Learner does not have to show this step. (5)

1.2 1.2.1

$$m_{PQ} = \frac{2-(-4)}{-4-(-2)} = \frac{6}{-2} = -3 \checkmark$$

$$m_{PR} = \frac{-1(-4)}{7-(-2)} = \frac{3}{9} = \frac{1}{3} \checkmark$$

$$m_{PQ} \times m_{PR} = -3 \times \frac{1}{3} = -1 \checkmark$$

$$\therefore PQ \perp PR \checkmark^{½}$$

$$\therefore \Delta PQR \text{ is a right angled triangle } \checkmark^{½}$$

(4)

1.2.2

$$PQ = \sqrt{(x_Q - x_P)^2 + (y_Q - y_P)^2}$$
$$= \sqrt{[-4-(-2)]^2 + [2-(-4)]^2} \checkmark$$
$$= \sqrt{4+36}$$
$$= \sqrt{40}$$
$$= 2\sqrt{10} \checkmark$$

$$PR = \sqrt{(x_R - x_P)^2 + (y_R - y_P)^2}$$
$$= \sqrt{[7-(-2)]^2 + [-1-(-4)]^2} \checkmark$$
$$= \sqrt{81+9}$$
$$= \sqrt{90}$$
$$= 3\sqrt{10} \checkmark$$

No marks for intermediary steps

$$A_{\Delta PQR} = \frac{1}{2} PQ \times PR$$
$$= \frac{1}{2} \times 2\sqrt{10} \times 3\sqrt{10} \checkmark^{½}$$
$$= 30 \ u^2 \checkmark^{½}$$

(5)

1.2.3

$$\tan\theta = -3 \checkmark^{½}$$
$$\therefore \theta = 180° - 71,565° \checkmark^{½}$$
$$= 108,435° \checkmark$$

Learner could have shown reference angle using positive value.

(2)

1.2.4

$$m = \left(\frac{x_Q + x_R}{2}; \frac{y_Q + y_R}{2} \right)$$
$$= \left(\frac{-4+7}{2}; \frac{2-1}{2} \right) \checkmark$$
$$= \left(1\frac{1}{2}; \frac{1}{2} \right) \checkmark$$

(2)

1.2.5 $m_{PR} = \frac{1}{3}$

$\therefore m_{MN} = \frac{1}{3}$ (PR $\perp$ MN)✓

$y - y_1 = m(x - x_1)$

$y - \frac{1}{2} = \frac{1}{3}(x - 1\frac{1}{2})$ ✓½

$y - \frac{1}{2} = \frac{1}{3}x - \frac{1}{2}$ ✓½

$\therefore y = \frac{1}{3}x$✓

(3)
[25]

QUESTION 2

2.1 $$\frac{\sin 135°.\sin 315°.\tan 225°}{\cos 330°.\sin 120°}$$

$$= \frac{\sin(180° - 45°).\sin(360 - 45°).\tan(180° + 45°)}{\cos(360° - 30°).\sin(180° - 60°)}$$✓ Accept any reduction.

$$= \frac{\sin 45°.-\sin 45°.\tan 45°}{\cos 30°.\sin 60°}$$✓

$$= \frac{\frac{\sqrt{2}}{2} \times \frac{-\sqrt{2}}{2} \times 1}{\frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2}}$$✓

$$= \frac{\frac{-2}{4}}{\frac{3}{4}}$$✓

$$= -\frac{2}{3}$$✓

(5)

2.2 $\dfrac{2\sin^2 x}{2\tan x - 2\sin x\cos x} = \dfrac{\cos x}{\sin x}$

LHS

$= \dfrac{2(1-\cos^2 x)}{2.\dfrac{\sin x}{\cos x}\checkmark - 2\sin x\cos x}$

$= \dfrac{2(1-\cos^2 x)}{\dfrac{2\sin x(1-\cos^2 x)}{\cos x}\checkmark\checkmark}$

$= \dfrac{2(1-\cos^2 x)}{1} \times \dfrac{\cos x}{2\sin x(1-\cos x)}\checkmark$

$= \dfrac{\cos x}{\sin x}\checkmark$

= RHS

Accept any other valid solution.
e.g.
LHS

$= \dfrac{2\sin^2 x}{2\dfrac{\sin x}{\cos x}\checkmark - 2\sin x\cos x}$

$= \dfrac{2\sin^2 x}{\dfrac{2\sin x - 2\sin x\cos^2 x}{\cos x}\checkmark}$

$= \dfrac{2\sin^2 x}{2\sin x(1-\cos^2 x)\checkmark} \div \dfrac{1}{\cos x}$

$= \dfrac{2\sin^2 x}{2\sin x(\sin^2 x)\checkmark} \times \dfrac{\cos x}{1}$

$= \dfrac{\cos x}{\sin x}\checkmark$

=RHS

(5)

2.3 $\dfrac{\tan^2 x + 1}{4} = 1$

$\tan^2 x + 1 = 4$

$\tan^2 x = 3\checkmark$

$\tan x = \pm\sqrt{3}$ ✓½

$\tan x = \sqrt{3}$ or $\tan x = -\sqrt{3}$

$x = 60°$ ✓½ $x = -60°$ ✓½

$x = 180° + 60° = 240°$ ✓½ $x = 360° - 60° = 300°$ ✓½

$x = 180° - 60° = 120°$ ✓½

$x \in (60°, 120°, 240°, 300°)\checkmark$

Or for easier marking

$\tan^2 x = 3$ ✓

✓ ✓ ✓ ✓

$x \in (60°; 120°; 240°; 3$

General solutions are not in the syllabus, but are acceptable as . (5)

2.4 2.4.1 $PR^2 = PQ^2 + RQ^2 - 2PQ.RQ\cos\hat{Q}$

$= 5^2 + 3^2 - 2(5)(3)\cos 120°$ ✓

$= 25 + 9 - 30\cos 120°$ ✓

$= 49$ ✓

$PR = 7$ m ✓

(4)

2.4.2 $\frac{\sin \hat{S}}{PR} = \frac{\sin P\hat{R}S}{PS}$

$\frac{\sin \hat{S}}{7} = \frac{\sin 38,2°}{5}$ ✓

$\sin \hat{S} = \frac{7 \sin 38,2°}{5}$ ✓

$= 60°$ ✓

Accept $59,971°$ (3)

2.4.3 $A = \frac{1}{2}(PQ)(QR)\sin \hat{Q}$ ✓

$= \frac{1}{2}(5)(3)\sin 120°$ ✓

$= 6,5 m^2$ ✓

Alternate

$s = \sqrt{s(s-a)(s-b)(s-c)}$

$s = \frac{a+b+c}{2} = 7,5$ ✓

Where

$A = \sqrt{7,5(7,5-5)(7,5-3)(7,5-7)}$ ✓

$A = 6,495 m^2$ ✓ (3)

[25]

QUESTION 3

3.1 3.1.1

16	48	57	60	62	62	62	66	67	68
69	70	70	70	76	78	80	80	88	160

$Q_{2\,pos} = \frac{2}{4}(20+1)$

$= 10,5$ ✓

$Q_2 = 68 + 0,5(69-68)$

$= 68,5$ ✓

or $\frac{68+69}{2}$ ✓ $= 68,5$ ✓ (2)

3.1.2

16	48	57	60	62	62	62	66	67	68
69	70	70	70	76	78	80	80	88	160

$$Q_{1\,pos} = \frac{1}{4}(20+1)$$
$$= 5,25 \checkmark$$
$$Q_1 = 62 + 0,25(62-62)$$
$$= 62 \checkmark$$
$$Q_{3\,pos} = \frac{3}{4}(20+1)$$
$$= 15,75 \checkmark$$
$$Q_3 = 76 + 0,75(78-76)$$
$$= 77,5 \checkmark$$

(4)

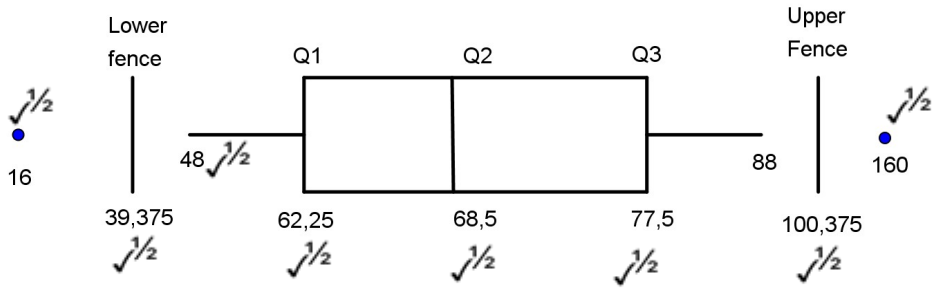
3.1.3 Upper fence $= 77,5 + 1,5(15,5)$ ✓
$= 100,25$ ✓

Use values form QUESTION 3.1.2 to get IQR.
CA marks awarded

Lower fence $= 62 - 1,5(15,5)$ ✓
$= 38,75$ ✓

(4)

3.1.4



(½ mark for each correct value) (4)

3.2 3.2.1

Classes (Data usage)	Frequency (f_i)	Midpoint (x_i)	$f_i \times x_i$	< Cumulative frequency
$0 \le x < 10$	7	5	35	7
$10 \le x < 20$	35	15	525	42
$20 \le x < 30$	12	25	**300** ✓½	**54** ✓½
$30 \le x < 40$	18	35	**630** ✓½	**72** ✓½
$40 \le x < 50$	5	45	**225** ✓½	**77** ✓½
$50 \le x < 60$	3	55	**165** ✓½	**80** ✓½

✓

(4)

TOTAL:	80		$\sum f_i x_i = 18$ 80	

3.2.2 $\frac{\sum f_i x_i}{n} = \frac{1880}{80}$ ✓

$= 23,5$ ✓ (2)

3.2.3 $Mo = l + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} \times c$

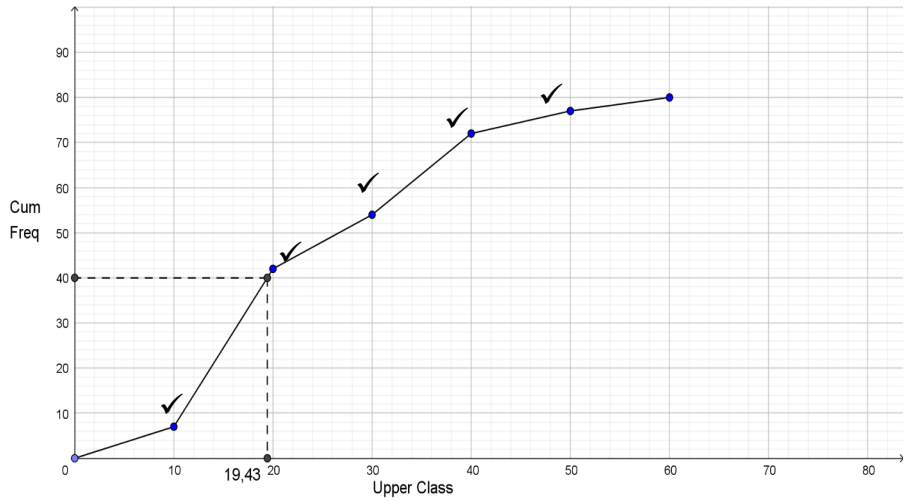
Award 3 marks if learner has chosen the incorrect class, but has substituted correctly.

$= 10$ ✓ $+ \frac{35-7}{2(35)-7-12} \times 10$ ✓

$= 10 + \frac{28}{51} \times 10$

$= 15,49$ ✓ (3)

3.2.4



(5)

3.2.5 Median = 19,43 ✓
(Accept answer from 18, but <20)

Learner must show how the median was obtained to get 1 mark. ✓ (2)

[30]

QUESTION 4

4.1

4.1.1	R19 000
4.1.2	+R1 000
4.1.3	
4.1.4	-R1 000
4.1.5	+R7 500
4.1.6	
4.1.7	R4 300
	R1700

-R2 600

(7 × 1) (7)

4.2

$$A = A_0(1+\frac{r}{100\times m})^{t\times m}$$

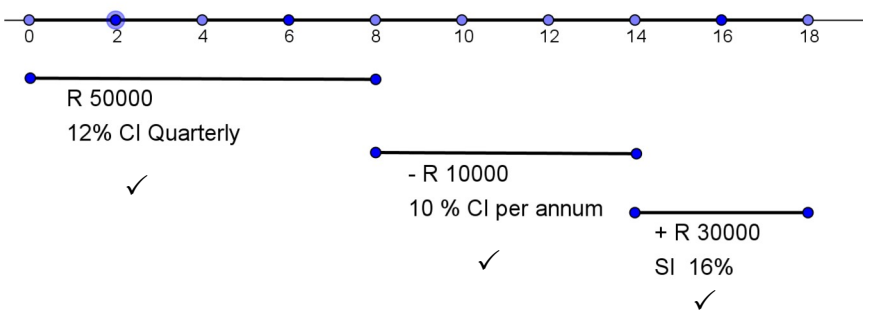
$$5000 = A_0(1+\frac{10,25}{100\times 4})^{8} \checkmark$$

$$P = \frac{5000}{(1+\frac{10,25}{400})^{8}} \checkmark$$

$$= 4083,77 \checkmark$$

(3)

4.3 4.3.1



(3)

4.3.2

$$A = A_0(1+\frac{r}{100\times m})^{t\times m}$$

$$= 50000(1+\frac{12}{100\times 4})^{32} \checkmark$$

$$= R128754,14 \checkmark$$

$$A_{8\,years} = R128754,14 - R10000 = R118754,14 \checkmark^{1/2}$$

$$A = P(1+i)^n$$

$$= 118754,14(1+0,1)^6 \checkmark$$

$$= R210380,20 \checkmark$$

$$A_{14\,years} = R210380,20 + R30000 = R240380,20 \checkmark^{1/2}$$

$\checkmark$

$$A = P(1+in)$$

$$= 240380,20(1+0,16\times 4) \checkmark$$

$$= R394223,53 \checkmark$$

(7)
[20]

TOTAL: 100

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